

DEVOPS MINI-PROJECT · CONTINUOUS DELIVERY

Jenkins–Docker Continuous Deployment

Community Health Survey System

Week 12 · Versioned image, registry, fresh container

Project: Community Health Survey System (CHSS)

Report date: 8 October 2026

Branch baseline: main · c9d29f68

Platform: Windows Jenkins · Docker Desktop · Java 17

Execution status: Jenkins pipeline changes and local Docker checks are implemented. Jenkins credential metadata returned HTTP 403, so Docker Hub credentials could not be confirmed. No Week 12 Jenkins push/deployment run is claimed.

01 — OBJECTIVE · 02 — REQUIREMENT · 03 — BASELINE

Goal and starting point

1. Objective

Extend the existing CHSS Jenkins pipeline so successful build and Selenium results gate a build-numbered Docker image push and a fresh container deployment.

2. Week 12 requirement

Git checkout → Maven build → Selenium tests → JUnit reports → versioned Docker image → Docker Hub push → fresh container → container verification → HTTP health check.

3. Existing system baseline

- Spring Boot 3.2.5, Java 17, Maven, executable WAR, existing multi-stage Dockerfile.
- Jenkins supports GIT_REVISION, publishes JUnit/artifacts, persists Selenium status, and gates Package/Deploy.
- Existing Tomcat deployment remains in the Jenkinsfile.
- Prior Jenkins build #16 is a green Week 10 baseline only, not Week 12 registry evidence.

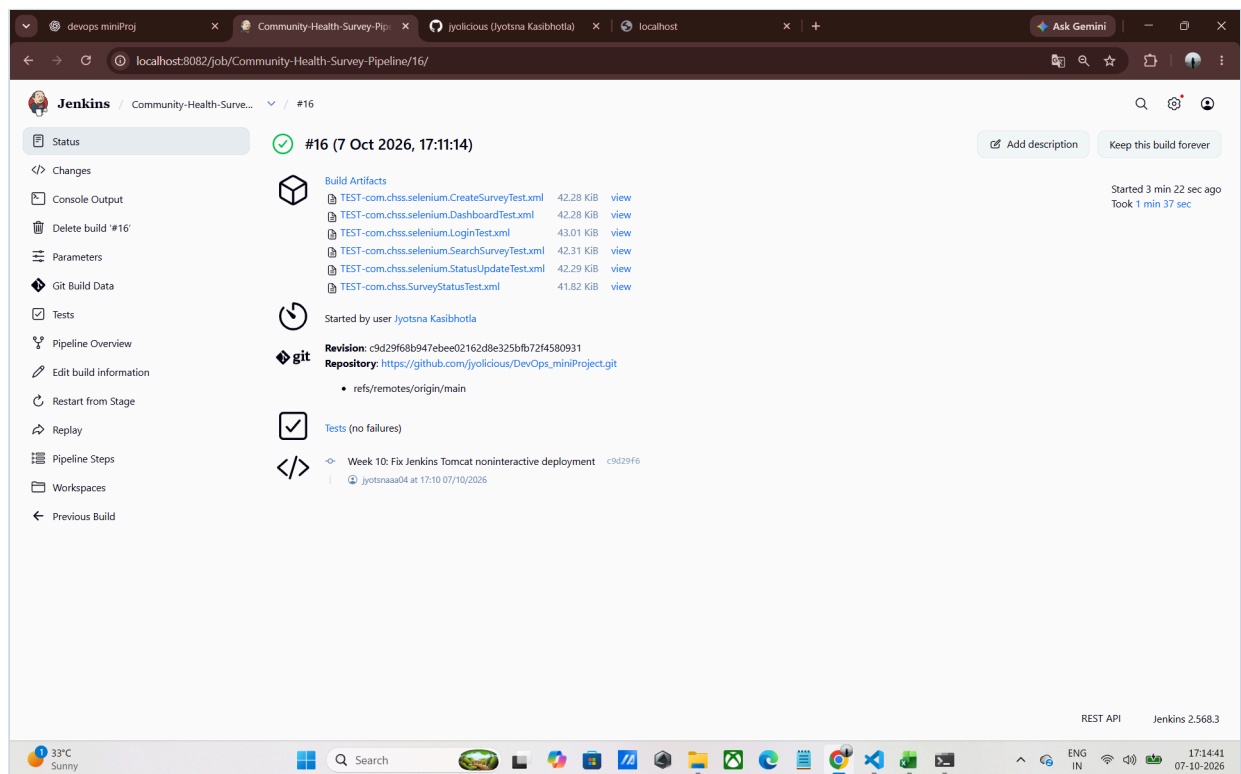
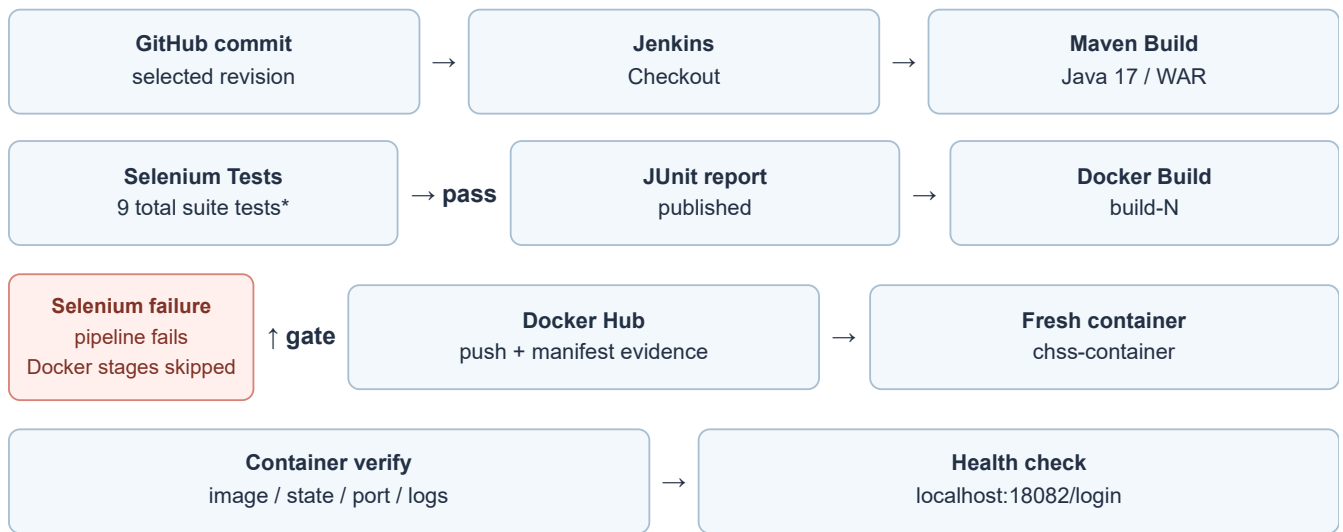


Figure 1. User-provided Jenkins build #16 baseline: green prior pipeline and no reported test failures. This predates Week 12 Docker stages.

The Docker stages are appended to the existing successful workflow; Week 10 checkout, test gate, reports and Tomcat deployment are retained.

04 — ARCHITECTURE

Commit-to-container design



*Existing Surefire reports contain 9 total tests: 6 Selenium UI tests and 3 survey-status unit tests. The Week 12 pipeline has not re-executed this suite.

The existing Publish Test Reports stage fails on any non-zero Selenium exit status. Jenkins stops subsequent stages after that error, so Docker Build, Login, Push and deployment cannot run after a failing test stage.

05 — JENKINS PIPELINE · 06 — SELENIUM GATE

Extend, preserve, gate

The existing Declarative pipeline is extended rather than replaced. Revision selection, Maven steps, Selenium app start/stop, persisted exit status, JUnit reports, screenshots, Package and Tomcat Deploy remain.

Existing stage	Week 12 treatment
Checkout / Build	Preserved; selected revision supported.
Start Test Application / Selenium Tests	Preserved. No Selenium files changed.
Publish Test Reports	Preserved. Non-zero saved test status fails before downstream stages.
Package / Tomcat Deploy	Preserved; Week 10 deployment remains.
Docker Build → Login → Push → Manifest Verify → Fresh Deploy → Container Verify → Health	Added sequentially; a failure stops later Docker operations.

```
Existing test gate remains:
if (seleniumExitCode != 0) {
    error("Selenium/Maven tests failed; refusing to package or deploy.")
}

Docker image pattern:
<credential username>/chss-app:build-<BUILD_NUMBER>
```

No Docker build/push is claimed after a failing test. No Week 12 Jenkins run has been triggered.

07 — IMAGE VERSIONING · 08 — DOCKER BUILD

Build-number image identity

Image name comes from the username in Jenkins credential `dockerhub-credentials` and the runtime Jenkins build number:

```
<DOCKERHUB_USERNAME>/chss-app:build-<BUILD_NUMBER>
```

- No build number is hardcoded.
- The version tag is the primary deployment reference; latest is not deployed.
- The existing Dockerfile is reused with the CHSS directory as build context.
- The exact versioned tag is pushed, verified in Docker Hub, pulled, and used for the fresh container.

```

REPOSITORY=chss-app TAG=week12-local-check ID=545938604f6a SIZE=474MB
REPOSITORY=chss-app TAG=week11 ID=95cd872e317b SIZE=474MB
REPOSITORY=chss-app TAG=week11-lifecycle ID=95cd872e317b SIZE=474MB
ImageID=sha256:545938604f6a2dba5194587d632bdb3ce7039cd8cf9136f8e815f8fb68b60a4f Created=2026-10-07T15:43:31.394814619Z RepoTags=["chss-app:week12-local-check"]
Container verification:
Running Docker containers:
ID=5699ada1be0f IMAGE=chss-app:week12-local-check STATUS=Up 5 minutes PORTS=0.0.0.0:18082->8081/tcp, [::]:18082->8081/tcp NAMES=chss-container
All CHSS deployment containers (including stopped):
ID=5699ada1be0f IMAGE=chss-app:week12-local-check STATUS=Up 5 minutes PORTS=0.0.0.0:18082->8081/tcp, [::]:18082->8081/tcp NAMES=chss-container
Container ID: 5699ada1be0f72490a7db38d4ca73cfaad6f4ec67b4dbcece391b06d0df028c7
Image: chss-app:week12-local-check
Running: true
Port bindings: {"8081/tcp":[{"HostIp":"0.0.0.0","HostPort":"18082"}, {"HostIp":"","HostPort":"18082"}]}
Application startup log milestones:
2026-10-07T19:13:40.452Z INFO 1 --- [community-health-survey] [main] c.chss.CommunityHealthSurveyApplication : Starting CommunityHealthSurveyApplication v0.1.0 using Java 17.0.20.1 with PID 1 (/app/app.war started by chss in /app)
2026-10-07T19:13:41.869Z INFO 1 --- [community-health-survey] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-10-07T19:13:44.178Z INFO 1 --- [community-health-survey] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path ''
2026-10-07T19:13:44.200Z INFO 1 --- [community-health-survey] [main] c.chss.CommunityHealthSurveyApplication : Started CommunityHealthSurveyApplication in 4.228 seconds (process running for 4.881)
Container verification passed.
Application health check:
Application health check passed: HTTP 200 at http://localhost:18082/login
  
```

Figure 2. Actual local-only image build/tag and container preflight. `week12-local-check` was not pushed and is not a Jenkins release tag.

Local Docker build passed. Local image ID:

`sha256:545938604f6a2dba5194587d632bdb3ce7039cd8cf9136f8e815f8fb68b60a4f`. The local-only tag was removed after validation.

09 — DOCKER HUB · 10 — AUTHENTICATION · 11 — PUSH

Registry authentication and evidence

Repository format: <Docker Hub username>/chss-app.

Jenkins expects credential ID `dockerhub-credentials`, type Username with password. The password field holds the project owner's Docker Hub PAT. The login helper sends it through standard input; the token is never embedded in source or command text.

Push and remote verification

1. Bind the Jenkins credential in Docker Login.
2. Authenticate in a per-build temporary Docker config directory.
3. Push <username>/chss-app:build-<BUILD_NUMBER>.
4. Inspect the remote Docker Hub manifest and archive the JSON result in Jenkins.
5. Delete the temporary Docker config after the pipeline.

Registry evidence pending: username, repository, build tag, push digest and registry screenshot are not available. Jenkins credential metadata returned HTTP 403. Docker Hub repository existence is also unverified.

If the repository does not exist under the credential username, create `chss-app` in Docker Hub before the push.

References: [Docker CLI login and password-stdin](#); [Jenkins Credentials Binding](#).

12 — FRESH CONTAINER DEPLOYMENT

Replace only the CHSS deployment

The Windows PowerShell deployment script pulls the just-pushed version before touching the old container. It checks the exact container name and confirms that its image or CHSS application label identifies it as CHSS.



Setting	Value
Container	chss-container
Host / container	18082:8081
Image	<username>/chss-app:build-<BUILD_NUMBER>
Database for lifecycle	In-memory H2 via runtime variables
Labels	CHSS application and Week 12 Jenkins management

The script refuses an unrecognized same-name container or another running container using port 18082. The old CHSS container remains if image pull or safety checks fail.

MySQL is not modified or claimed as tested. H2 supports container lifecycle validation without database credentials.

13 — CONTAINER VERIFICATION · 14 — HEALTH CHECK

Verify the application, not just docker run

Container verification waits up to 90 seconds for application startup, then checks that:

- The container exists and is running.
- The configured image matches the build-number image.
- Host port 18082 maps to container port 8081.
- The CHSS deployment label is present.
- Logs show Spring Boot/Tomcat readiness.

A separate step polls `http://localhost:18082/login`, follows redirects, and succeeds only for HTTP 2xx. An unreachable endpoint or non-2xx response fails the build.

```

Week 12 Local Preflight Eviden
REPOSITORY=chss-app TAG=week12-local-check ID=545938604f6a SIZE=474MB
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Container verification:
Running Docker containers:
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Container ID: 56999adalbe0f72490a7db38d4ca73cfaad6f4ec67b4dbcece391b06d0df028c7
Image: chss-app:week12-local-check
Running: true
Port bindings: {"8081/tcp":[{"HostIp":"0.0.0.0","HostPort":"18082"},{"HostIp":"","HostPort":"18082"}]}
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2026-10-07T19:13:41.869Z INFO 1 --- [community-health-survey] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 808
1 (http)
2026-10-07T19:13:44.178Z INFO 1 --- [community-health-survey] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path ''
2026-10-07T19:13:44.200Z INFO 1 --- [community-health-survey] [main] c.chss.CommunityHealthSurveyApplication : Started CommunityHealthSurveyApp
lication in 4.228 seconds (process running for 4.881)
Container verification passed.
Application health check:
Application health check passed: HTTP 200 at http://localhost:18082/login

```

Figure 3. Actual local container state, 18082→8081 mapping, startup logs and HTTP 200.

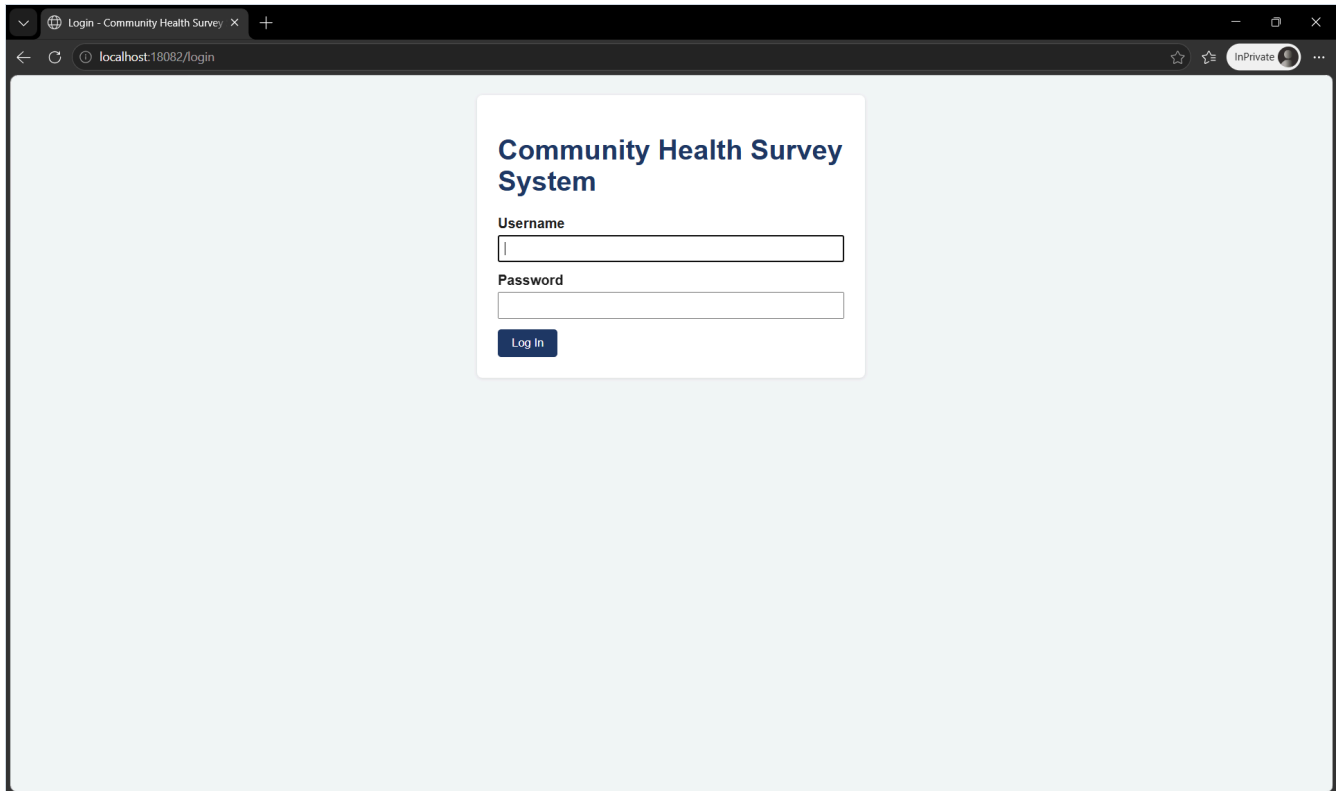


Figure 4. CHSS login page served at localhost:18082 using the local validation image.

These prove local Docker behavior only; they do not show a Docker Hub pull or Jenkins-created container.

15 — END-TO-END COMMIT-TO-CONTAINER EVIDENCE

Evidence available and pending

Requested proof	Current status
Git commit	Source baseline c9d29f68; Week 12 commit pending final review.
Jenkins / Maven / Selenium / JUnit	Prior Week 10 build #16 screenshot is green; Week 12 not triggered.
Docker build / image details	Local-only build passed; captured.
Docker Hub repository / tag / push	Not available; credential/repository unconfirmed.
Fresh Jenkins container / inspect / logs	Not available; local preflight was verified and removed.
Application health	Local-only HTTP 200 at port 18082; browser screenshot captured.
Final Week 12 Jenkins success	Pending credential verification and one run.

No Jenkins build number, Docker Hub tag, digest, push result, or remote registry screenshot has been fabricated.

16 — SECURITY AND CREDENTIALS

Keep authentication outside the repository

- Docker Hub authentication uses Jenkins Credentials ID `dockerhub-credentials`.
- Credential type is Username with password; the password value is a Docker Hub PAT entered by the owner in Jenkins.
- The login helper reads the masked environment and passes the PAT to `--password-stdin`.
- The PAT is not written to Jenkinsfile, scripts, Dockerfile, README, PDF or Git, and is not echoed to Jenkins logs.
- Docker auth config is limited to a temporary per-build directory and removed in post `{ always }`.
- No database password is included in build arguments, Dockerfile, report or runtime demonstration.
- Only username/repository and build tag are logged.

Credential presence and permissions must be verified by an authorized Jenkins user. No token was requested or accessed here.

Manual Jenkins credential setup

Jenkins → Manage Jenkins → Credentials → System → Global credentials (unrestricted) → Add Credentials. Choose **Username with password**, enter Docker Hub username and PAT, set ID to `dockerhub-credentials`, then save. Keep the PAT out of chat and source control.

Fail closed and report the boundary

Failure	Behavior
Maven / Selenium	Reports publish; non-zero Selenium status fails before Docker build.
Docker build / login	Later push and deployment stages are skipped.
Push / remote manifest	Deployment is skipped; old container is untouched.
Image pull / replacement safety	Old CHSS container remains untouched.
Wrong image, stopped container, wrong port, missing startup log	Container verification fails with selected metadata/log diagnostics.
Unreachable / non-2xx endpoint	Health check fails pipeline.

What remains

1. Verify/add the Jenkins credential with Docker Hub write access.
2. Confirm/create repository <username>/chss-app.
3. Run the parameterized job once against main.
4. Capture real Jenkins, registry, pushed tag/digest, container and health evidence.

Jenkins responds at localhost:8082, but job and credential REST endpoints return HTTP 403. That prevented credential inspection and the Week 12 run. The Jenkins pipeline linter was also inaccessible without authentication.

Docker Hub username and repository existence are unknown and are not guessed here.

19 — CONCLUSION

Implementation status

The existing Jenkins pipeline now contains the Week 12 Docker continuous-deployment path while preserving the Week 10 Selenium gate and Week 11 Dockerfile.

Local validation confirmed the existing Dockerfile builds, a fresh CHSS container runs on 18082→8081, startup logs are correct, and the app returns HTTP 200. The health check also fails when no service listens. The temporary local container and image tag were removed.

The Docker Hub login, push, remote manifest, Jenkins replacement container and final Jenkins success are not yet validated. They require confirming the Jenkins credential and repository, then running the job once.

Database limitation: MySQL connectivity from the Docker container was not verified during Week 12. The local lifecycle validation used H2 to verify container startup and health without changing the existing MySQL database.

20 — FILES MODIFIED

Change inventory

Path	Purpose
community-health-survey/chss/JenkinsFile	Docker build, secure login, push, manifest evidence, safe deploy, container and health verification.
community-health-survey/chss/scripts/dockerhub-login.ps1	PAT over stdin.
community-health-survey/chss/scripts/deploy-docker.ps1	Pull version, validate target, replace CHSS container.
community-health-survey/chss/scripts/verify-docker-container.ps1	Image/state/port/label/log verification.
community-health-survey/chss/scripts/verify-docker-health.ps1	Bounded HTTP 2xx health check.
community-health-survey/chss/scripts/verify-docker-registry.ps1	Remote manifest evidence artifact.
community-health-survey/chss/Dockerfile, .dockerignore	Existing Week 11 files; added to Git because Jenkins checkout needs them.
community-health-survey/chss/README.md	Concise Week 12 flow/security note.
community-health-survey/chss/WEEK11_DOCKER_LIFECYCLE.md	Correct actual Week 11 image/container metadata.
documentations/Week12_Jenkins_Docker_Continuous_Deployment.pdf	This report.
documentations/week12-evidence/*.png	Two actual local preflight screenshots.

No Jenkins configuration, Selenium tests, application source, database configuration or credentials were changed. Existing unrelated working-tree files are excluded.

21 — VALIDATION RESULTS

Observed outcomes

Check	Result
PowerShell parser	Pass for all five new scripts.
Docker build	Pass locally; temporary tag <code>chss-app:week12-local-check</code> .
Local container	Pass; ID prefix <code>5699ada1be0f</code> .
Port mapping	Pass; <code>18082→8081/tcp</code> .
Startup logs / container verify	Pass; Spring Boot/Tomcat and safe metadata verified.
Local health check	Pass; HTTP 200 at <code>/login</code> .
Unreachable health check	Pass; helper returned exit code 1 after container removal.
Cleanup	Pass; temporary container/image tag removed, Week 11 image tags retained.
Jenkinsfile linter	Not run; Jenkins REST endpoints return 403; no local Groovy compiler.
Docker Hub login/push/manifest	Not run; credential presence unavailable.
Week 12 Jenkins Selenium/JUnit/build/deploy	Not run; no new Jenkins build triggered.
Git commit / push	Pending final review.

Local Docker implementation checks are complete. End-to-end registry/Jenkins evidence remains pending an authorized credential check and one Jenkins run.